over the first year and the unwind values of the remaining trades that we have on. Recall that when we first entered into the trade, we received in 2s at 5.11% and we paid in 5s at 5.60%. Also note that 1-year LIBOR at the time was 5.00%, since the 1-year swap rate was given as 5.00%. The carry in the first year from receiving on \$538.37mm 2s is[18]

$$\$538{,}368{,}293 \times (5.11\% - 5.00\%) = \$592{,}205.$$

The carry in the first year from paying on \$233.57mm 5s is

$$\$233{,}574{,}545 \times (5.00\% - 5.60\%) = -\$1{,}401{,}447.$$

Thus, the net carry in the first year is $\$592{,}205 - \$1{,}401{,}447 = -\$809{,}242$. Now we need to compute our mark-to-market on the remaining trade. Remember, although we originally paid in 5s and received in 2s, a year has gone by and now our current position has us paying in 4s at 5.60% and receiving in 1s at 5.11%. As shown in Figure 8.9, the value of the 4-year leg is \$164,013 and the value of the 1-year leg is \$51,224, both in our favor, giving a total value of \$215,237. The total mark-to-market on the trade is the value of the carry plus the value of the remaining position, or $-\$809{,}242 + \$215{,}237 = -\$594{,}005$. So we put on a 2s5s steepener at 49 bps. After a year, 1s4s, the relevant benchmark, is 52 bps. The curve steepened, but we still lost money. How can that be?

In order to get a handle on why we lost money even though the curve steepened, let's go back to our original curve in Table 8.1 and see what the forwards have to say. We can compute forward swap rates effective in one year from this curve, and these are displayed in Table 8.4. Remember, we can loosely regard forward rates as the market's expectation of where rates will be in the future. At a minimum, they are rates that we can definitively lock in by trading forward starting swaps. So, for example, the market expects that the 1-year swap rate in one year will be 5.226% and expects that the 4-year swap rate in one year will be 5.772%. Note that the difference between these two rates, $5.772\% - 5.226\% = 54.6$ bps, is what the market expects 1s4s will be in a year.

So we put on a 2s5s steepener at 49 bps. After a year, 1s4s (the benchmark we should look at) was 52 bps. The curve steepened, but we lost money because it didn't steepen enough. The forwards indicate that the market was expecting the 1s4s curve to steepen more than the 52 bps that we wound up with. The market was telling us at the time of trade inception that 1s4s was expected to be 54.6 bps in a year, but we only got 52 bps. That's why we lost money.[19]

[18] For ease of exposition, we are displaying rounded notional amounts throughout the text. However, all calculations are based on the exact notional amounts.

[19] As mentioned previously, another issue to contend with in this trade is that it does not remain duration-neutral over time. Whereas the notionals of the 2-year and 5-year legs of the trade where chosen so as to be duration-neutral, the residual 1s4s trade we are

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.100%	5.100%	(13,080,175)	(12,445,456)	11,912,302	11,334,255
2	5.233%	5.367%	(13,080,175)	(11,811,534)	12,535,892	11,320,041
3	5.412%	5.769%	(13,080,175)	(11,167,265)	13,475,536	11,504,807
4	5.649%	6.362%	(13,080,175)	(10,499,272)	14,860,652	11,928,437
5	5.789%	6.355%	0	0	0	0
				(45,923,527)		46,087,540
		Fixed rate	5.600%	NPV	164,013	
		Notional	233,574,545			

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.100%	5.100%	27,510,620	26,175,661	(27,456,783)	(26,124,437)
2	5.233%	5.367%	0	0	0	0
3	5.412%	5.769%	0	0	0	0
4	5.649%	6.362%	0	0	0	0
5	5.789%	6.355%	0	0	0	0
				26,175,661		(26,124,437)
		Fixed rate	5.110%	NPV	51,224	
		Notional	538,368,293			

Figure 8.9: Computing the Value of the 2s5s Steepener After One Year Based on the Curve Shown in Table 8.3

Let's take a step back and think about the following strategy: suppose we put on a curve trade and

1. Earn carry (either positive or negative) in the first year
2. Unwind the remaining position at the end of the first year.

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left with after a year is not. As a practical matter, investors will often adjust the notional on one or both of the legs (by paying or receiving in additional swaps) so as to maintain a duration-neutral position.

Forward Swap	Rate
1x2	5.226%
1x3	5.462%
1x4	5.644%
1x5	5.772%

Table 8.4: Forward Swap Rates Effective in One Year Based on the Curve in Table 8.1

Maturity	Swap Rate
1 Year	5.226%
2 Year	5.462%
3 Year	5.644%
4 Year	5.772%

Table 8.5: The Swap Curve in One Year Assuming that the Forwards in Table 8.4 Are Realized

Our expected profits (as determined by the forwards) from this strategy should be zero. That is to say, if we put on a curve trade and earn carry in the first year, and the *forwards are realized* come the end of the first year, then our profit will be zero.[20] To see that this is the case, suppose we put on our 2s5s steepener today. As we computed previously, our carry over the first year is −\$809,242. Now assume that the forwards are realized in one year, so that the swap curve in one year is as shown in Table 8.5. As shown in Figure 8.10, the mark-to-market of the remaining 1s4s trade assuming the forwards are realized is exactly \$1,401,447 − \$592,205 = \$809,242. Thus, assuming the forwards are realized, our profit is zero. We put on a steepener and the market expected the curve to steepen. We earned negative carry in the first year (intuitively, we "had to pay" carry in the first year since we were putting on a steepener when the market expected the curve to steepen; if we had instead put on a flattener, we would have been going against the forwards and would have "gotten paid," that is, earned positive carry, in the first year

[20]To say that the *forwards are realized* in one year means that in one year from now the prevailing swap curve is the one suggested by the forwards. That is, given our example, it means that the prevailing swap curve in one year is the one given in Table 8.4, with the 1-year swap rate at 5.226%, the 2-year swap rate at 5.462%, and so on. The fact that this strategy must result in zero profits if the forwards are realized is a direct consequence of Equation 8.1.

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.226%	5.226%	(13,080,175)	(12,430,584)	12,206,018	11,599,839
2	5.469%	5.712%	(13,080,175)	(11,758,892)	13,342,251	11,994,495
3	5.659%	6.041%	(13,080,175)	(11,088,967)	14,111,089	11,962,945
4	5.795%	6.202%	(13,080,175)	(10,441,383)	14,486,496	11,563,994
				(45,719,825)		47,121,273
		Fixed rate	5.600%	NPV	1,401,447	
		Notional	233,574,545			

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.226%	5.226%	27,510,620	26,144,380	(28,133,772)	(26,736,585)
2	5.469%	5.712%	0	0	0	0
3	5.659%	6.041%	0	0	0	0
4	5.795%	6.202%	0	0	0	0
				26,144,380		(26,736,585)
		Fixed rate	5.110%	NPV	(592,205)	
		Notional	538,368,293			

Figure 8.10: The Mark-to-Market of the Steepener in One Year Assuming the Forwards Are Realized

for doing so). When the curve did indeed steepen at the end of the first year in accordance with the forwards, the mark-to-market gain on our remaining position was just enough to offset our negative carry, giving us zero profit.

Carry and roll-down analysis is a common tool for looking at curve trades as well. Table 8.6 shows carry and roll-down numbers for a variety of curve trades in basis points running. It also shows a tabular representation of the carry and roll-down numbers for outright swaps from the perspective of the fixed rate receiver taken from Figure 8.6.

In computing the carry and roll-down numbers for curve trades, we assume that we receive on the shorter-tenor swap and pay on the longer-tenor swap on PV01 weighted notionals (i.e., that we enter into a swap curve steepener).

Swaps			
	Carry	Roll-Down	Total
2-year swap	11.6	11.0	22.6
3-year swap	16.2	19.0	35.2
4-year swap	17.4	17.0	34.4
5-year swap	17.2	13.0	30.2
Curve Steepeners			
	Carry	Roll-Down	Total
2s3s	−4.6	−8.0	−12.6
2s4s	−5.9	−6.0	−11.9
2s5s	−5.6	−2.0	−7.6
3s4s	−1.2	2.0	0.8
3s5s	−0.9	6.0	5.1
4s5s	0.3	4.0	4.3

Table 8.6: Carry and Roll-Down on Swaps of Different Maturities (Reflective of Data in Figure 8.6) and of Curve Steepeners, One Year Horizon, Based on the Swap Curve Given in Table 8.1

So what do these numbers mean? Suppose, for example, we enter into a 2s5s curve steepener and choose notionals such that we are duration neutral. This trade will experience −7.6 bps of carry and roll-down over a year. In other words, if we enter into the steepener and if after a year the residual swap curve has not steepened at least 7.6 bps, our mark-to market on the trade will be negative.[21]

Sometimes people will talk about a trade they really love (from a fundamental or technical perspective) but may make mention that it carries really badly. For example, suppose an investor really likes the 2s5s steepening trade, believing that sector of the curve is likely to steepen out. When he puts the trade on, he doesn't know if the curve will steepen today, tomorrow, in a month from now, or longer. But in general he has conviction that the curve will steepen and wants to put the trade on now so that he doesn't miss out on the opportunity. But every day he is in the trade and the curve doesn't steepen costs him money.

[21] Of course, if we had entered into a flattener, these numbers would be reversed, and under the assumption that the curve had not moved after a year, our mark-to-market would be in our favor.

If he is somewhat flexible as to what points on the curve he wants to initiate a steepener, these carry and roll-down numbers suggest that he should look for a different trade. From Table 8.6 we see that all of the steepeners involving a 2-year swap have poor carry and roll-down characteristics.[22] He might be tempted to put on a 3s5s steepener instead, which has a relatively attractive carry and roll-down profile, owing to the high carry and roll-down profile of 3-year swaps.

8.2.2 Forward Yield Curve Trades

Sometimes people look to exploit opportunities for curve trades in *forward space*.[23] Let's say, for instance, that we think that the 1s3s curve (referring here to 1-year swaps versus 3-year swaps, not a 1s3s basis swap) is going to steepen. Moreover, we may think that either fundamental or technical forces will cause 1s3s to remain steep into the foreseeable future. This means that we think that for some considerable period of time each day when we come in and look at the screens we will see 1s3s steeper than it is today. Given our view, we may be interested in looking at forward curves to see if we can find an attractive opportunity for putting on a steepener.

Let's go back to our original swap curve, as shown in Table 8.1. Note that 1s3s is currently at $5.30\% - 5.00\% = 30$ bps. So we can get into a spot starting 1s3s steepener at 30 bps, but there may be opportunities to put on the trade at more attractive levels if we enter into a *forward curve trade*, a curve trade using forward starting swaps. Let's look two years out. We can compute forward swap rates effective in two years from our swap curve in Table 8.1, as shown in Table 8.7. Notice that if we put on a 1s3s steepener two years forward, we can get in at the more attractive level of $5.976\% - 5.712\% = 26.4$ bps.[24] That is, we can improve our entry level from the spot level of 30 bps by 3.6 bps. Let's assume that we want to put on this 1s3s steepener two years forward for \$100,000 per bp. As shown in Figure 8.11, we need to receive on \$1,167.98mm of the 2x3, and we need to pay on \$412.57mm of the 2x5 swap.

Suppose that we hold this trade for two years, and after two years we were proved wrong and the curve did not steepen at all. Assume in fact that

[22] This is because 2-year swaps, as shown in this figure, have relatively poor carry and roll-down characteristics as compared to other points on the curve.

[23] *Space* has become an industry buzzword of the new millennium. We have Treasury space, mortgage space, agency space, LIBOR space (a fancy way of saying swap space), and so on. A good definition for space in this context is the span of securities or tradeable instruments within a particular asset class or subset of an asset class.

[24] If we are putting on a steepener, we benefit when the curve differential is greater than our entry point. Thus a low entry point is advantageous.

Forward Swap	Rate
2x3	5.712%
2x4	5.872%
2x5	5.976%

Table 8.7: Forward Swap Rates in Two Years Based on the Spot Curve in Table 8.1

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	0	0	0	0
2	5.113%	5.226%	0	0	0	0
3	5.312%	5.712%	24,694,470	21,142,838	23,566,903	20,177,441
4	5.494%	6.041%	24,694,470	19,938,293	24,924,931	20,124,366
5	5.635%	6.202%	24,694,470	18,773,918	25,588,026	19,453,242
				59,855,048		59,755,048
		Fixed rate	5.986%	NPV	100,000	
		Notional	412,571,222			

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	0	0	0	0
2	5.113%	5.226%	0	0	0	0
3	5.312%	5.712%	66,834,340	57,222,025	66,717,542	57,122,025
4	5.494%	6.041%	0	0	0	0
5	5.635%	6.202%	0	0	0	0
				57,222,025		57,122,025
		Fixed rate	5.722%	NPV	100,000	
		Notional	1,167,982,789			

Figure 8.11: Computing the Notionals on Each Leg of the Forward Steepener for a Duration-Neutral Trade with a PV01 of $100,000

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	(24,653,213)	(23,479,251)	20,628,561	19,646,249
2	5.113%	5.226%	(24,653,213)	(22,313,218)	21,559,934	19,513,542
3	5.312%	5.712%	(24,653,213)	(21,107,514)	23,566,903	20,177,441
4	5.494%	6.041%	0	0	0	0
5	5.635%	6.202%	0	0	0	0
				(66,899,983)		59,337,231
		Fixed rate	5.976%	NPV	(7,562,752)	
		Notional	412,571,222			

Time	zc	Forward	Fixed	PV Fixed	Float	PV Float
1	5.000%	5.000%	66,717,542	63,540,516	(58,399,139)	(55,618,228)
2	5.113%	5.226%	0	0	0	0
3	5.312%	5.712%	0	0	0	0
4	5.494%	6.041%	0	0	0	0
5	5.635%	6.202%	0	0	0	0
				63,540,516		(55,618,228)
		Fixed rate	5.712%	NPV	7,922,288	
		Notional	1,167,982,789			

Figure 8.12: Computing the Value of the Forward Steepener After Two Years Based on the Curve Shown in Table 8.1

the curve two years later is exactly the same as it was when we got into the trade, as depicted in Table 8.1. Let's look at the P&L on this trade. The first thing to note is that there was no carry on this trade during the first two years since both of these trades were forward starting in two years. In fact, the trade that we put on two years ago is now simply a position in spot starting 1-year and 3-year swaps. As shown in Figure 8.12, the P&L of the trade after two years is \$7,922,288 − \$7,562,752 = \$359,536. Note that despite the fact that the curve did not steepen at all, we still made money by entering into the forward steepener and taking advantage of the better entry level.[25]

8.2.3 Conditional Yield Curve Trades

We saw in Section 8.2.2 that people sometimes enter into forward curve trades in order to improve their entry point from levels obtainable from the spot curve. Sometimes people will look to use options to further enhance the entry level that they can obtain in a yield curve trade. The primary motivation behind entering such trades, referred to generically as *conditional yield curve trades*, is a view on the curve. However, using options introduces other risks as well.

Let's go back to our original swap curve as shown in Table 8.1, and let's suppose that we are still interested in putting on the 1s3s steepener, two years forward. Recall from Table 8.7 that the 2x3 swap rate is 5.712% and the 2x5 swap rate is 5.976%, meaning we can get in the forward steepener at $5.976\% - 5.712\% = 26.4$ bps. Suppose further that swaption vols for at-the-money swaptions are as shown in Table 8.8. Now consider the following trade:

1. Buy $412.57mm of a 2 into 3 payer swaption struck at 5.976%
2. Sell $1,167.98mm of a 2 into 1 payer swaption struck at 5.712%.

If both of these trades finish in-the-money and are exercised in two years, then we will be paying in 3s and receiving in 1s from the counterparty who exercised against us. That is, conditional on both options being exercised, we will be in the 1s3s forward steepener that we discussed before. This

[25] Recall from Chapter 3 that a forward starting swap can be decomposed into two spot starting swaps. For instance, receiving on a 2x3 is equivalent to receiving on a 3-year spot starting swap and paying on a 2-year spot starting swap of the same notional. Similarly, paying on a 2x5 is equivalent to paying on a spot starting 5-year swap and receiving on a spot starting 2-year swap, both with the same notional. If we decompose our forward curve trade into these four components, we will find that we have a bunch of carry in the first two years (and all LIBOR legs cancel out). And after two years we will be receiving for a year at a rate of 5.30% on $1,167.98mm and paying for three years at a rate of 5.60% on $412.57mm, which is a difference of 30 bps. Recall that in the forward curve trade we were able to get into the trade at 26.4 bps. Referring again to the decomposition of our forward curve trade into four components, can you guess the sign of the carry in aggregate over the first two years?

The carry must be positive over the first two years. Think about it this way: we can earn some carry over the first two years and get into a forward 1s3s steepener in two years at a (relatively poor) level of 30 bps, or we can just get into a forward 1s3s steepener in two years at a (relatively good) level of 26.4 bps. We should be indifferent between these two trades. Thus, the carry in the first two years afforded by the decomposition into four spot starting swaps must be positive.

Swaption	Lognormal Vol	Normal Vol
2 into 1	18%	102.5 bps
2 into 3	16%	95.4 bps

Table 8.8: Swaption Vols for At-the-Money Swaptions

type of conditional curve trade is referred to as a *bearish steepener*.[26] Note that, conditional on exercise, our entry point is still 5.976% – 5.712% = 26.4 bps. However, as shown in Figure 8.13, we will get paid to enter into this trade. The \$412.57mm 2 into 3 payer costs us \$5,382,623, and we sell the \$1,167.98mm 2 into 1 payer for \$5,785,353, so we wind up pocketing \$402,730. So by using options we were able to get into a steepener trade, conditional on both options finishing in-the-money, and we were able to take out an upfront as well. We were able to do this because we exploited the difference between implied volatility on 2 into 1 swaptions and 2 into 3 swaptions, buying the (relatively) cheaper 2 into 3 swaption vol and selling the (relatively) more expensive 2 into 1 swaption vol.[27]

It is important to note, however, that this conditional curve trade does not exhibit the same view as the forward curve trade we analyzed in Section 8.2.2. The view being taken in this trade is that the 1s3s curve will steepen in two years *conditional on a market sell-off*. Now that may jibe very well with our view. Or we may not be interested in this trade at all. We may have put on the outright forward steepener and really have in the back of our mind that the market was more likely to steepen in a rally and more likely to flatten in a sell-off.

[26] The word *bearish* denotes that this steepener uses payer swaptions that will finish in-the-money if the market sells off. Note that these options are both struck at-the-money forward, although swap rates will have to sell off from their current (i.e., spot) levels in order for the payer swaptions to finish in-the-money.

[27] Table 8.8 presents implied vols in both lognormal and normal terms. Given our discussion in Chapter 5, it is probably more informative in general to look at the normal vols in order to gauge the relative richness or cheapness of swaption vol. In this particular case, we get the same answer either way.

What if instead we were to consider taking advantage of the vol differential by structuring a trade using receiver swaptions? In this case we would want to sell 2 into 1 year receiver swaptions and buy 2 into 3 receiver swaptions. These options would finish in-the-money, conditional on the market rallying. Assuming both options finish in-the-money, we would be receiving in a 3-year swap and paying in a 1-year swap. Thus, conditional on a rally, we would be in a flattener. This type of trade is known as a *bullish flattener*. If someone had a flattening view, this type of trade might be an attractive alternative that would provide a better entry point than would be obtained by putting on a forward flattener.